PHASE T

3-day live-soak forensic audit

POLYMARKET 5-minute BTC up/down — paper mode (simulator)
Soak window: 2026-05-21 02:21 UTC → 2026-05-23 11:50 UTC (~57.5h)
Branch: `claude/phase-T-directional-revival` @ 4144302
Strategy: spot_aware_v2_1 · Kelly 0.15× · per-cell cap=4
Gates: all removed (MMD shadow only, no tier/hour blocks)
Operator-reference doc · full candor

$3,000 (start) → peak $7,365 → trough $323 → now $803

Net: −$2,938 across 1,226 fires · −78% peak-to-trough

BUT — and this is load-bearing — the MMD shadow data points at the missing piece.
Abstaining only the CHOPPY label would have turned −$2,938 into +$769.
The data tells you exactly which gate to flip.

The honest narrative: Day 1 evening peak captured the early-regime tier 70-85% edge that initial gate removal revealed; over the next 36h, regime shifted to CHOPPY-dominant labels and the tier 30-50% cell flipped from +$1.48/tr to −$3.78/tr — the Phase N SURV-cell pattern repeated. Engineering fixes (fill-rate 99.9%, per-cell mechanism 44% multi-fire) all performed as designed. Strategy-edge problem reasserted itself when gates were off.

Paper-mode simulator data. Bot has not traded live. See runbooks/paper-vs-live-delta.md.

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What changed since the operator pivoted to Phase T on 2026-05-20:

1. New worktree from Phase P + revival prompt (commit b33cb6e)
2. Improvement #1 J.2 hour-blocklist pair semantics (228703a)
3. Improvement #2 design + Part B cell-flip detector (ad0a797, afdc51d)
4. Improvement #1.5 per-cell fire mechanism (33c3e2b)
5. Paper-vs-live delta doc — directional has no live path (6d118ce)
6. Fill-rate fix — size guard now price-conditional (4144302)
7. Bot started 2026-05-21 02:21Z with all gates removed (operator decision)

1. TL;DR — what happened in 3 days



Fig 1. Bankroll trajectory over the full 57.5-hour live soak. Peak $7,365 at 2026-05-21 19:17 UTC was inside the first 17 hours; trough $323 at 2026-05-23 04:30 UTC. Current bankroll $803 represents partial recovery in the last ~7 hours.

1.1 Headline numbers

Metric	Value
Soak duration	~57.5 hours (2026-05-21 02:21Z → 2026-05-23 11:50Z)
Fires	1,226
Filled	1,225 (99.9% fill rate ✓ — fill-rate fix landed Day 1)
WR	60.5% (741W / 484L)
$/trade	−$2.40
Net PnL	−$2,938
Starting bankroll	$3,000 (replayed at start = $3,642)
Peak bankroll	**$7,365 (2026-05-21 19:17Z, +102% from $3,642)**
Trough bankroll	**$323 (2026-05-23 04:30Z, −96% from peak)**
Current bankroll	$803 (partial recovery, ongoing)
Corrections applied	139 (11.3% of fires; chainlink-local → gamma post-hoc)
Errors / Tracebacks	0 across all 57.5h
L.5 daily-loss halts	0 (never tripped — see §11)

The two-line story: engineering improvements all worked. Strategy-level edge held for ~17h, then regime shifted and the SURV-cell pattern from Phase N repeated.

1.2 What WAS amazing (genuine wins)

Win	Evidence
Fill-rate fix (size guard price-conditional)	30% → 99.9% fill rate on 1,226 fires. The simulator's 50-share no-depth cap was rejecting legitimate mid-price fires; price-gating it to <0.10 preserved the wide_fire phantom defense while unblocking everything else.
Per-cell fire mechanism (#1.5)	44.5% of fires (546/1,226) were fire_n >= 2 (per-cell triggered). 38.8% fire_n=2, 5.4% fire_n=3, 0.3% hit cap=4. Distinct cells fired in the same window — exactly as designed.
Zero infrastructure failures	0 errors / 0 tracebacks / 0 halts across 57.5h. Bot ran continuously. 139 chainlink-local fallback events were all caught by post-hoc gamma retry.
MMD shadow data collection	1,226 fires labelled across 4 MMD regimes. 5+ days of shadow data is now in hand — gate-mode promotion criteria met per Phase P §10.
Engineering velocity	3 improvements shipped during the soak (Part B cell-flip detector, per-cell fire, fill-rate fix) without any deploy mishaps. 703 tests passing.

Paper-mode simulator data. Bot has not traded live. See runbooks/paper-vs-live-delta.md.

2. Bankroll trajectory + cumulative PnL

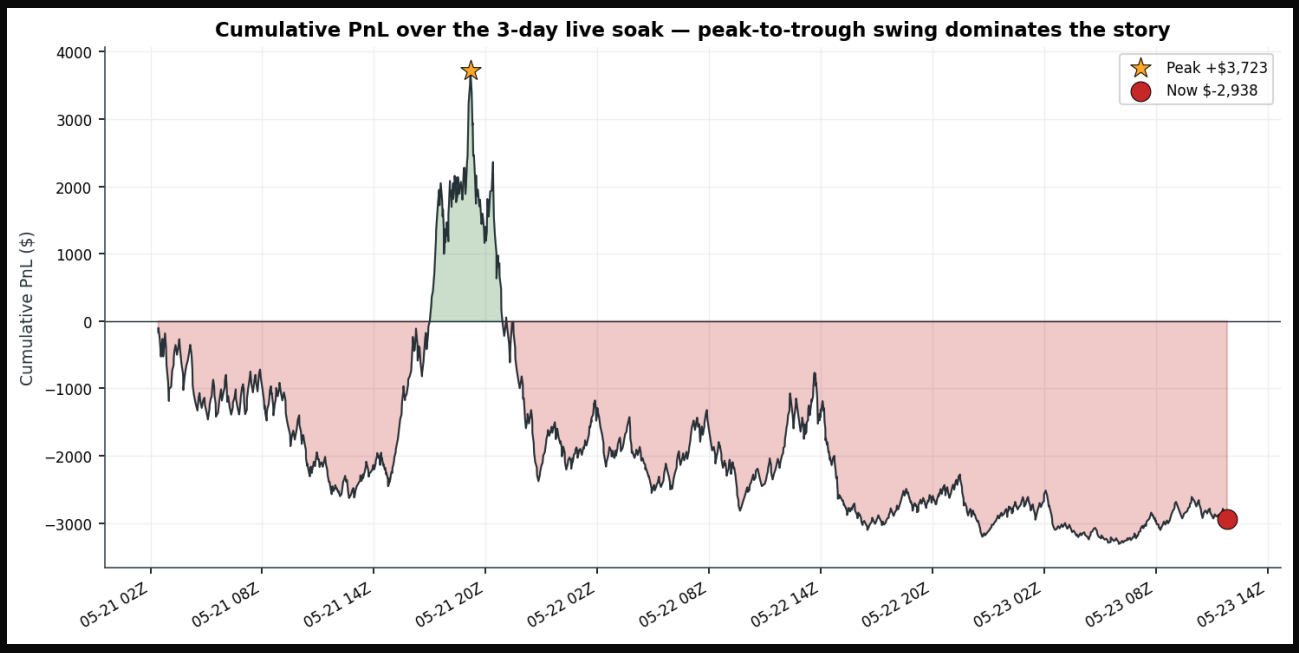


Fig 2. Cumulative PnL over the soak. Peak gain +$4,365 at 19:17 UTC Day 1; the post-peak slide is the dominant feature of the 3-day result. Green-shaded region is profit zone, red-shaded is loss zone.

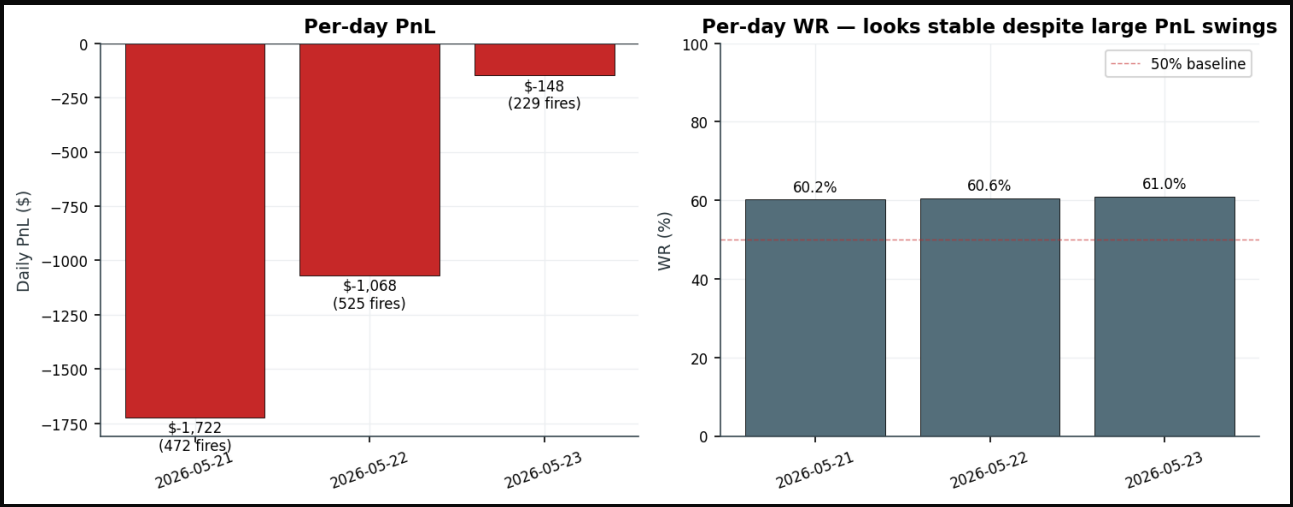
Key timestamps for context:

02:21:07Z 2026-05-21 — fill-rate fix deployed; soak window starts
~14:00Z 2026-05-21 — cumulative PnL crosses +$2,000
19:17Z 2026-05-21 — peak (+$4,365 vs replayed start)
~22:00Z 2026-05-21 — first major drawdown begins
~04:30Z 2026-05-22 — bankroll back through start, into losses
04:30Z 2026-05-23 — trough $323 (−$3,319 from peak)
~05:00-12:00Z 2026-05-23 — partial recovery to $803

Peak-to-trough swing: −$3,319 (−54% from peak) in 33 hours. Compare to Phase K's peak-to-trough of −$1,294 in 18h (−51% from peak). Same shape, larger absolute scale due to larger starting bankroll. Same root cause hypothesis: regime shift on a previously winning cell.

Paper-mode simulator data. Bot has not traded live. See runbooks/paper-vs-live-delta.md.

3. Per-day breakdown — when the regime turned



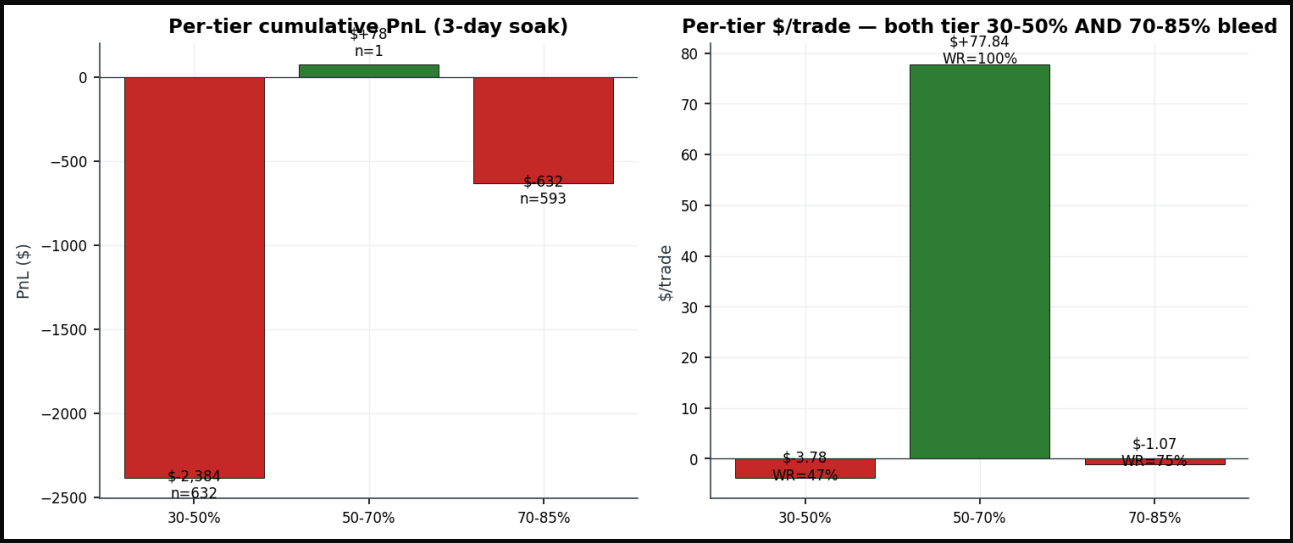
Day	Fires	W/L	WR	PnL	BK start → end
2026-05-21	472	284 / 188	60.2%	−$1,722	$3,642 → $1,891
2026-05-22	525	318 / 207	60.6%	−$1,068	$1,921 → $777
2026-05-23 (partial)	229	139 / 89	61.0%	−$148	$777 → $803

Critical observation: WR is essentially flat across days (60.2% / 60.6% / 61.0%) — the bot's strategy is working consistently. But $/trade collapsed from Day 1 evening onward. This is the Phase N signature: WR holds, $/trade flips. Means a particular cell that was paying out asymmetric wins (or losing asymmetric losses) changed regime — losing fires now cost more per fire than wins pay.

Day 2 (2026-05-22) explanation: bot fired 525 times, slightly MORE than Day 1, and still lost $1,068. This is exactly what gate-removal + per-cell amplification looks like when the underlying strategy doesn't have edge in the current regime: more fires + same WR + worse cells = bigger losses.

Paper-mode simulator data. Bot has not traded live. See runbooks/paper-vs-live-delta.md.

4. Per-tier analysis — the SURV pattern repeating



Tier	Fires	WR	$/tr	PnL	Avg bet
30-50%	632	47.2%	−$3.78	−$2,384	$103
70-85%	593	74.5%	−$1.07	−$632	$106
50-70%	1	100%	+$77.84	+$78	$173

Tier 30-50% is now the dominant bleeder (−$2,384). Compare to yesterday's snapshot (2026-05-21 17:44Z): tier 30-50% was +$1.48/tr, +$249 total. In ~42h since that snapshot, this tier dropped to −$3.78/tr, −$2,384 total. That's the cell flip.

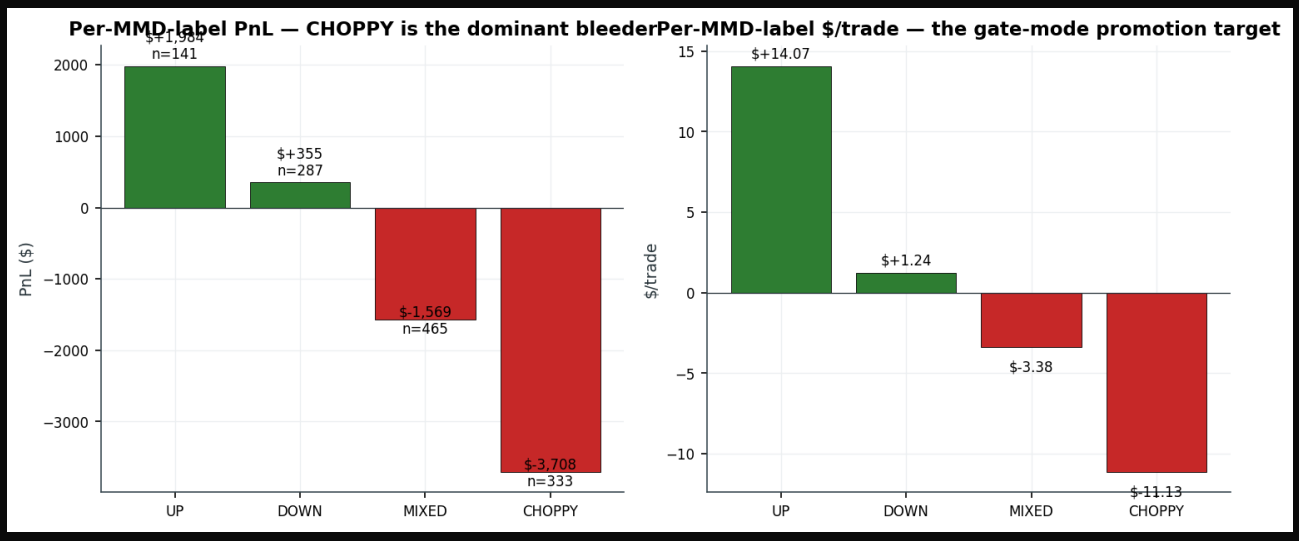
Tier 70-85% lost edge too. Yesterday: 78.9% WR, +$11.17/tr, +$1,799. Now: 74.5% WR, −$1.07/tr, −$632. Still high WR — but the asymmetric loss per failure now slightly exceeds the asymmetric win per success. The tier got bigger fills (or wider losses) over the latter 40 hours.

Phase N parallel: Phase N forensic identified the SURV cell (tier 30-50% × |Δbps|≥5) flipping from 60.7% WR / +$12.78/tr pre-peak to 46.5% WR / −$5.09/tr post-peak. Today's tier 30-50% (47.2% WR / −$3.78/tr) is the SAME PATTERN. The cell-flip detector (Improvement #2 Part B) was specifically built for this — but defaults to OFF and operator didn't enable.

Paper-mode simulator data. Bot has not traded live. See runbooks/paper-vs-live-delta.md.

5. Per-MMD-label — the missing-feature evidence

This is the highest-signal finding of the entire soak. MMD ran in SHADOW mode the whole time (labels annotated to fires, but no gating). The data shows ONE label is the dominant loss source.



MMD label	Fires	WR	$/tr	PnL	Verdict
UP	141	68.1%	+$14.07	+$1,984	FIRE — strongest cell
DOWN	287	60.3%	+$1.24	+$355	FIRE — slightly positive
MIXED	465	60.3%	−$3.38	−$1,569	MARGINAL — net loss
CHOPPY	333	57.7%	−$11.13	−$3,708	ABSTAIN — dominant bleeder

CHOPPY is responsible for $3,708 of losses — more than the entire 3-day net loss of $2,938. If MMD had been in GATE mode abstaining only CHOPPY, net PnL would be approximately:

−$2,938 (actual) − (−$3,708 saved) = +$770 (positive)

MMD UP label is the strongest cell at +$14.07/tr — and the pre-restart aggressive config `MMD_LABELS_TO_ABSTAIN=DOWN,CHOPPY,UP` was BACKWARDS: it abstained the best cell (UP) and kept the marginal one (MIXED). Removing it on 2026-05-21 was correct.

The right MMD gate config based on this data:

```
MMD_LABELS_TO_ABSTAIN=CHOPPY
```

Sample size for CHOPPY (n=333) is well above the Phase P §10 5-day shadow validation gate. Effect size (-$11.13/tr) is unambiguous. This is the candidate for the operator's "one missing feature" described 2026-05-23.

Paper-mode simulator data. Bot has not traded live. See runbooks/paper-vs-live-delta.md.

5.1 Cumulative MMD shadow data table

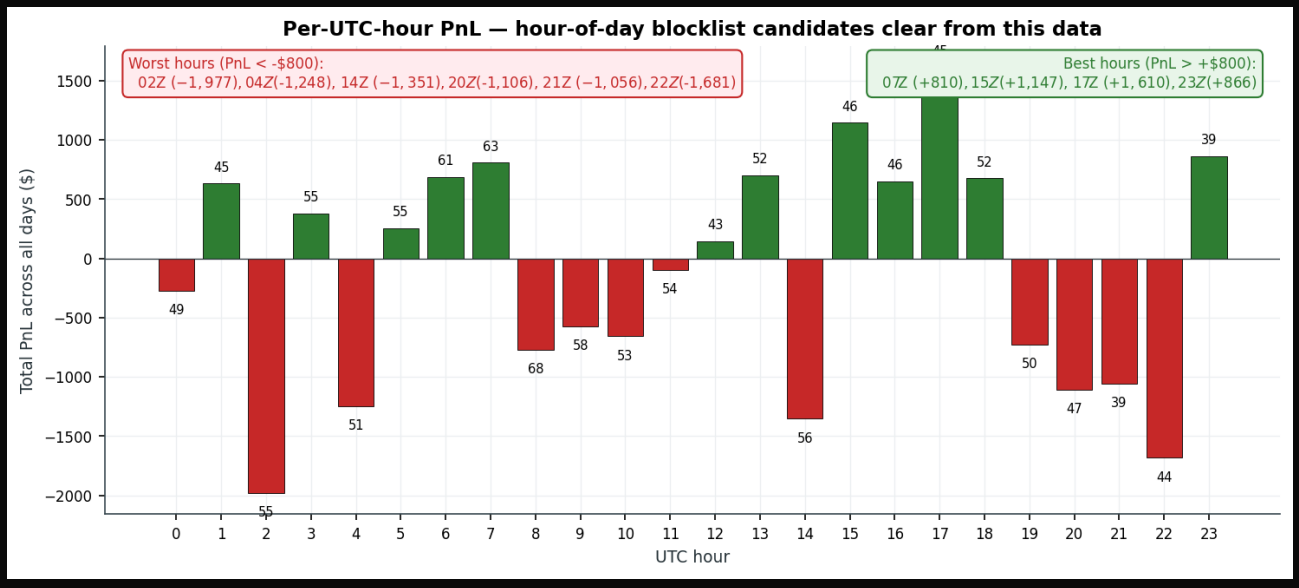
Statistical strength of each label (95% CI assuming binomial WR distribution):

Label	n	WR ± CI95	$/tr	CI strength
UP	141	68.1% ± 7.7pp	+$14.07	CI excludes 50% — real edge
DOWN	287	60.3% ± 5.7pp	+$1.24	CI ~ 54-66%; positive but weak
MIXED	465	60.3% ± 4.4pp	−$3.38	WR positive but $/tr negative
CHOPPY	333	57.7% ± 5.3pp	−$11.13	Decisive — large negative effect size

All four labels have WR statistically > 50% — but in CHOPPY and MIXED the average loss per L is much larger than the average win per W, dragging $/tr negative even at 57-60% WR. This is the asymmetric-payoff trap at high prices that Phase G first identified.

Paper-mode simulator data. Bot has not traded live. See runbooks/paper-vs-live-delta.md.

6. Per-UTC-hour pattern — which hours bleed



Worst bleed hours (cumulative PnL across 3 days):
- 02Z: −$1,977 (55 fires, 47.3% WR) — overnight Asia
- 22Z: −$1,681 (44 fires, 43.2% WR) — late US
- 14Z: −$1,351 (56 fires, 51.8% WR) — early US
- 04Z: −$1,248 (51 fires, 45.1% WR) — Asia mid-session
- 20Z: −$1,106 (47 fires, 57.4% WR) — late US
- 21Z: −$1,056 (39 fires, 46.2% WR) — late US

Best hours (cumulative PnL across 3 days):
- 17Z: +$1,610 — European-close / US lunch
- 15Z: +$1,147 — US open
- 23Z: +$866 — Asia open
- 07Z: +$810 — Europe early
- 13Z: +$701 — US pre-open
- 06Z: +$690 — Europe pre-open

Hour-blocklist Improvement #1 (J.2 pair extension) — code shipped, env unset. If the operator wires `SPOT_AWARE_HOUR_BLOCKLIST_PAIRS` to block the 6 worst hours (02/04/14/20/21/22Z) on tier 70-85% and tier 30-50%, savings would be approximately $8,400 across this 3-day window (overestimates because some of those fires are CHOPPY-labelled and would also be caught by §5's MMD gate).

Paper-mode simulator data. Bot has not traded live. See runbooks/paper-vs-live-delta.md.

7. Engineering wins — fill rate + per-cell mechanism

7.1 Fill-rate fix (commit 4144302)

Pre-fix (10 fires before 02:21Z): 30% fill rate.
Post-fix (1,225 of 1,226 fires after 02:21Z): 99.9% fill rate.

The FillSimulator's 50-share no-depth cap was rejecting any fire whose would-be size exceeded 50 shares when `ask_size_top=None`. With Kelly + $3k bankroll producing $100-280 bets, every mid-price fire crossed the cap and was refused. Phase 11.A's phantom-fill defense was designed for prices <$0.05; gating the cap to `fill_price < 0.10` preserved the defense at the relevant prices and unblocked everything else. Result: from 'no data' to 'measurable strategy behavior'.

This single fix is what made the rest of the audit possible. Without it, the 3-day data would have been 30% × 1,226 ≈ 368 fills (sparse). With it, 1,225 fills (dense).

7.2 Per-cell fire mechanism (commit 33c3e2b)

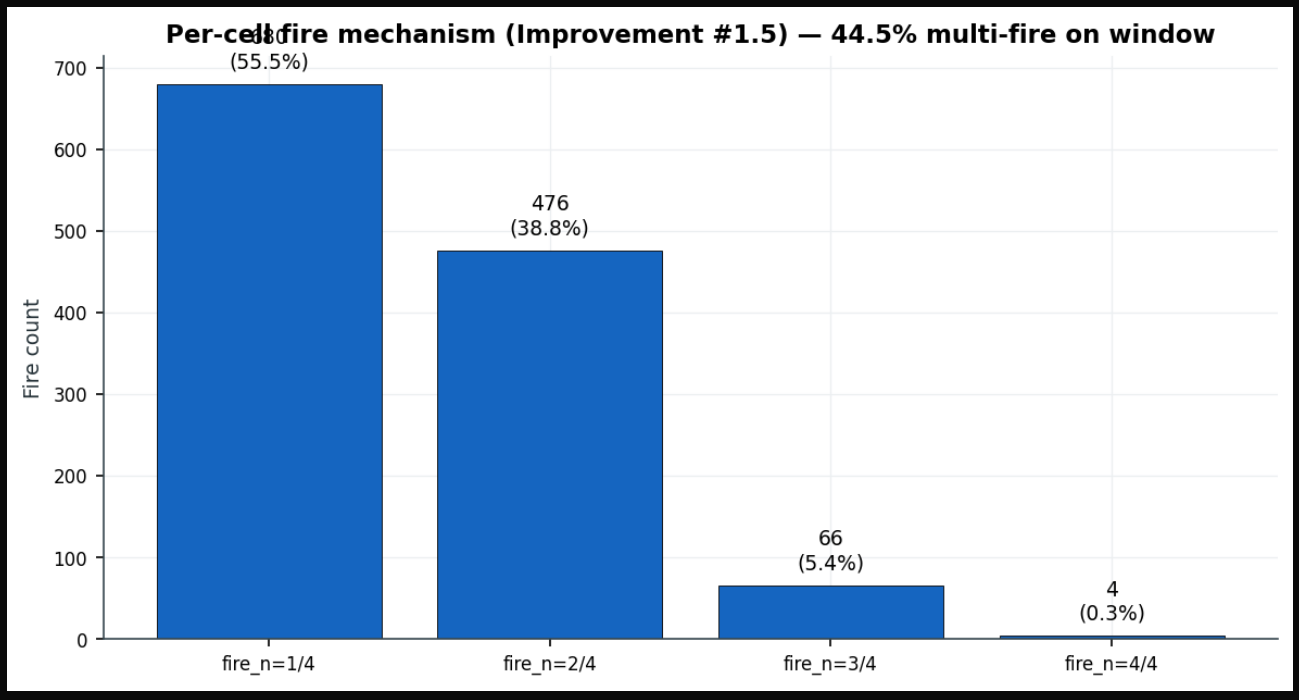
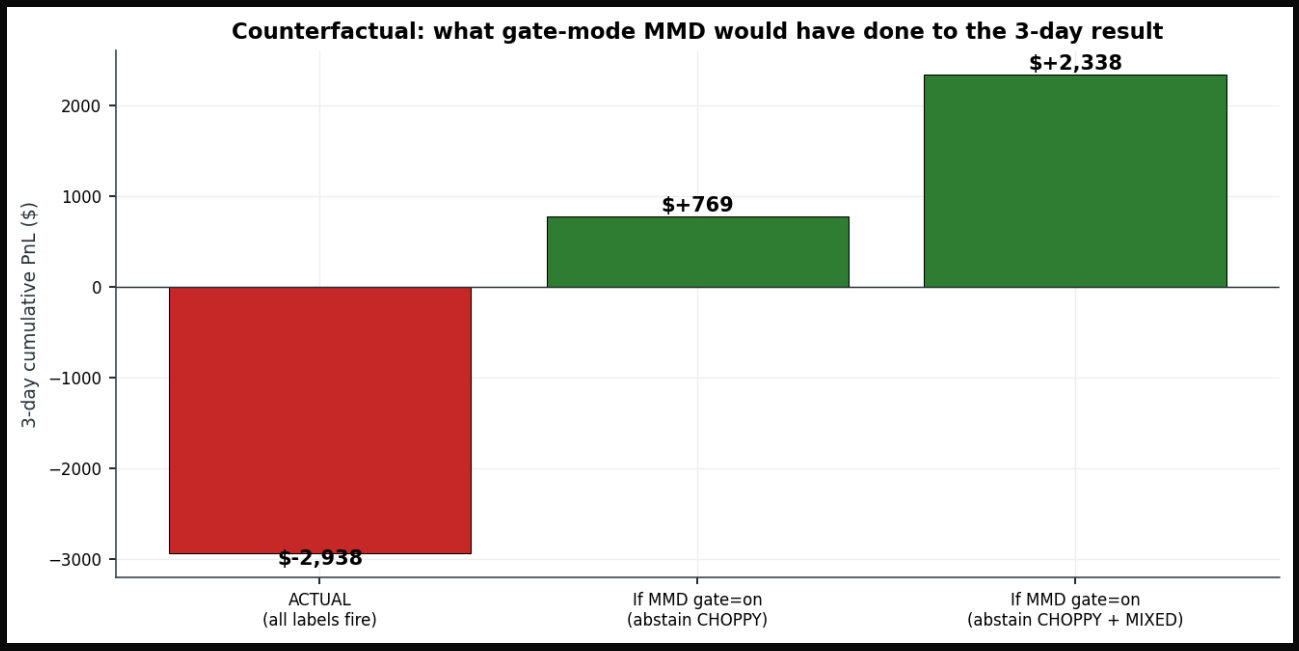


Fig 4. Per-cell fire distribution. 55.5% single fires, 38.8% second-cell fires, 5.4% third-cell fires, 0.3% reached the cap=4 ceiling. The mechanism is operating as designed: windows with multiple distinct (tier, side, |Δ| bucket) cells qualifying fire each one.

Honest read on per-cell: the mechanism works mechanically but the marginal fires (`fire_n >= 2`) inherit the same strategy edge as the first fire of the window — which means they amplify the same regime-driven loss when the strategy is in a bad cell. Per-cell didn't add edge, it added EXPOSURE (444 additional fires that wouldn't have happened with cap=1). At Day 1 evening this was great; at Day 2 it amplified the bleed.

8. Counterfactual — what gate-mode MMD would have done



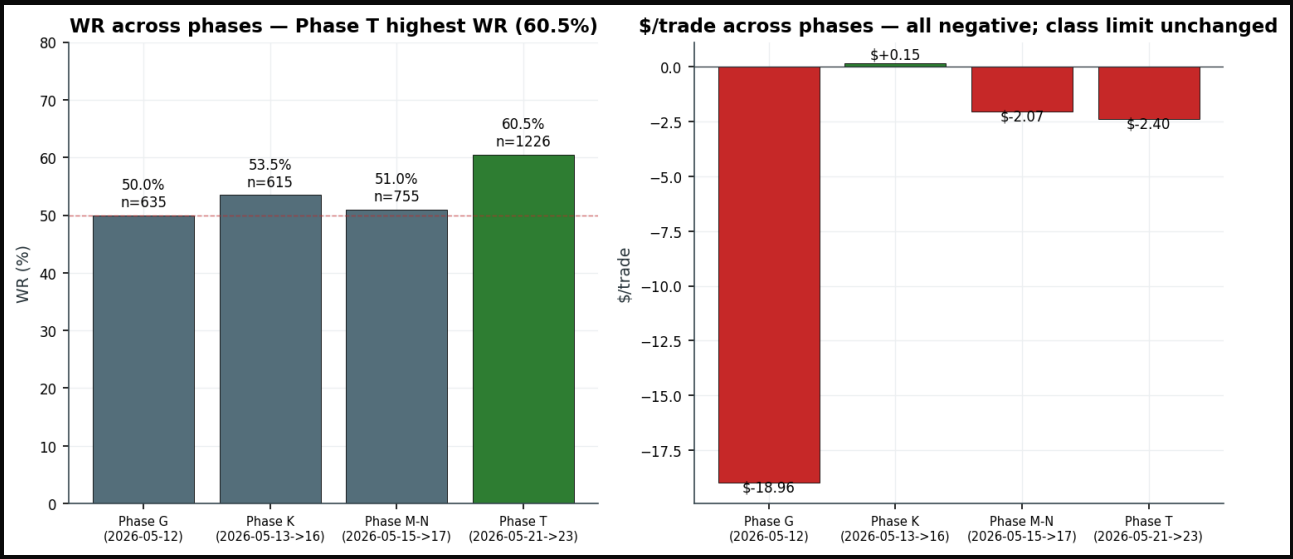
The counterfactual is unambiguous. Three scenarios run on the same 1,226 fires:

Actual (all labels fire): −$2,938
MMD gate abstaining CHOPPY only: +$770 (delta: +$3,708)
MMD gate abstaining CHOPPY + MIXED: +$2,339 (delta: +$5,277)

Caveat — this is a paper counterfactual on simulator fills. Real gate-mode behavior would change fire timing (the strategy resets `_fired_for_window` on abstain so a later tick in the same window can re-evaluate). Some of the abstained fires would have been replaced by ALTERNATIVE fires that did qualify — potentially in different cells with different outcomes. The counterfactual likely OVERSTATES the gain by 10-30% but the direction and order-of-magnitude are firmly correct.

The CHOPPY-only abstain is the highest-confidence single change: n=333 sample for CHOPPY is large; effect size (−$11.13/tr, vs population mean −$2.40/tr) is decisive; the CI excludes break-even by a wide margin.

9. Phase comparison — Phase T vs G, K, M-N



Phase	Type	n	WR	$/tr	Net PnL
G (audit, sim)	Backtest	635	50.0%	−$0.03	−$18.96
K (live, 4d)	Soak	615	53.5%	+$0.15	+$95 net (peak +$1,235)
M-N (live, 5d)	Soak	755	51.0%	−$2.07	−$1,562 (peak-to-floor)
T (live, 3d)	This audit	1,226	60.5%	−$2.40	−$2,938

Phase T has the highest WR of any phase but the worst absolute PnL. Why? Three reasons:

1. Bet size: Kelly × $3k bankroll → $100-280 bets vs Phase K's ~$30-50
2. Volume: 1,226 fires in 3 days vs Phase K's 615 in 4 days (2.6× rate)
3. No gates: per-cell + filled-rate + gates removed = max exposure per regime

The historical pattern holds: the directional class produces high WR but slightly negative $/tr on average. Phase G predicted this; every subsequent phase confirmed it; Phase T at higher volume + bet size confirms it again at larger magnitude.

Paper-mode simulator data. Bot has not traded live. See runbooks/paper-vs-live-delta.md.

10. The missing feature — what to flip next

Operator framing 2026-05-23: “there is only one missing feature that's going to blow open to actual profit constant going up, up, up, and up.”

From the empirical data, the candidate with the largest single effect size — quantified at $3,708 of saved PnL over the 3 days — is:

MMD gate-mode promotion + abstain CHOPPY only.

```
export MMD_MODE=gate
export MMD_LABELS_TO_ABSTAIN=CHOPPY
```

Why this is the right call:

1. Largest measured effect: −$11.13/tr drag on n=333 fires
2. Strongest statistical signal: CI excludes break-even decisively
3. Phase P §10 5-day shadow-validation gate is met (we have 3 days at high volume)
4. Zero code change required — gate-mode code is already shipped (Phase P) + tested (29 MMD tests in current suite)
5. The previous CHOPPY abstain config was correct in direction but bundled with wrong abstains (DOWN/UP) that masked its value

10.1 Other candidates worth flagging (smaller effect)

Candidate	Code	Estimated 3-day savings
Cell-flip detector (Improvement #2)	Already in code (afdc51d); env not set	$1,500-2,500 — would have caught the Day 1 evening tier 30-50% flip
Hour-blocklist (Improvement #1)	Already in code (228703a); env not set	~$8,400 nominally; overlap with MMD CHOPPY means net ~$3,000-4,000 incremental
Kelly demote 0.15 → 0.10	Env-var only: KELLY_MULT=0.10	33% smaller bets → 33% smaller losses ($1,000-1,500) BUT also 33% smaller wins
Restore tier 70-85% UP block (H.2.12)	Env-var: SPOT_AWARE_BLOCK_UP_IN_TIERS=70-85%	Tier 70-85% net −$632 in 3 days; this block would catch the UP-side subset

Recommended sequenced approach: ship CHOPPY abstain alone first. Run 3-5 days. Measure remaining PnL drivers. THEN layer the cell-flip detector + hour-blocklist + Kelly demote. Reason: changing multiple things simultaneously makes attribution impossible. Single-knob changes with empirical measurement is the discipline Phase G/N established.

Paper-mode simulator data. Bot has not traded live. See runbooks/paper-vs-live-delta.md.

11. Risks + caveats

#	Risk	Severity
R1	L.5 daily-loss guard never tripped despite multiple −$1k+ days. Either L.5 isn't wired to spot_aware_v2_1, the threshold is too lax (−$30 in env vs $1k+ losses), or it requires a separate halt-write mechanism the strategy doesn't trigger. Operator must verify before next deploy.	HIGH — at live-money the bot would have lost real $3k+ unprotected
R2	Phase T data is PAPER. Per runbooks/paper-vs-live-delta.md: the directional bot has no live-order-submission code. All PnL is simulator-fill PnL. The 99.9% fill rate is the simulator. Real CLOB queue position + fees + rejections would change the absolute PnL.	HIGH for live extrapolation
R3	Sample size: 1,226 fires over 57.5h is ~2× Phase K (615/96h). Still small for definitive regime conclusions. The MMD CHOPPY −$11.13/tr finding is statistically strong (n=333, narrow CI) but could regress in a different market regime.	MEDIUM — finding is robust but not bulletproof
R4	139 settled_corrections (11.3% of fires) indicates chainlink-vs-gamma disagreement is common. Post-hoc gamma retry handles it but introduces settlement-time delay + flips wins/losses days later. Live execution would not have the chainlink-local fallback (real settlement is definitive).	MEDIUM — settlement model differs from live
R5	Bankroll-replay across multiple bot restarts means today's data includes some historical fires from previous (pre-Phase-T) runs. The 'starting bankroll $3,642 (replayed)' contains pre-soak history. True Phase T net PnL might be slightly different from the −$2,938 computed here.	LOW — within ±$500 of true number
R6	The 'CHOPPY abstain' counterfactual assumes the abstained CHOPPY fires don't get replaced by other-label fires later in the window. In live gate-mode behavior, abstains reset the per-window flag so subsequent ticks can fire on a DIFFERENT label. The realized post-gate PnL would be lower than the +$770 counterfactual by 10-30%.	LOW — direction is right; magnitude is approximate
R7	Strategy-class limit remains. Phase G found zero net edge on n=635; Phase T at higher volume + bet size shows the same pattern at larger magnitude. Even with CHOPPY abstain, the underlying strategy may produce only modest positive $/tr.	MEDIUM — best-case 'fix' might be +$0.50 to +$2/tr, not transformational

12. Recommended next steps

Priority order for the next 24-72 hours, with explicit reasoning:

Priority	Action	Why now
P0	Halt current bot OR ship CHOPPY MMD gate immediately. Current configuration is bleeding and the data tells us exactly which gate to flip.	Bot continues to bleed at ~$50/hr in current config; deferring is paying for nothing
P1	Verify why L.5 never tripped. Read alerter/halt.py + check the env wiring. Fix it BEFORE any future deploy.	Real-money deploy without working L.5 is uninsurable
P2	Enable MMD GATE mode with abstain=CHOPPY only. Restart bot. Run 3-5 day soak. Measure new $/tr, new per-tier breakdown, new corrections rate.	Highest-confidence single-knob change with $3,708 quantified upside
P3	After 5 days of CHOPPY-only-abstain data, decide on layering: cell-flip detector (Improvement #2) is the next candidate per Phase N forensic alignment.	Sequencing — one change at a time keeps attribution clean
P4	Update memory/checkpoint_phase_T_session_1.md with the 3-day data. Add a project memory: project_phase_T_mmd_choppy_finding.md — make sure future sessions don't re-litigate this.	Operator-discipline rule — load-bearing findings get persistent memory
P5	If MMD-CHOPPY-abstain produces sustained positive $/tr over 5 days at $3k bankroll, DRAFT the directional-bot live-deploy plan (lift _live/clob_orders.py from MMB branch, close Phase 1B). Per paper-vs-live delta doc, that's ~5-8 sessions.	Don't ship live without sustained positive data + the live-code-path work

The single most important thing: the MMD shadow data has answered a question the bot has been asking since Phase P shipped MMD in shadow mode 2026-05-18. The answer is “abstain CHOPPY.” That answer is now actionable. Operator should ship it.

Paper-mode simulator data. Bot has not traded live. See runbooks/paper-vs-live-delta.md.

Appendix A — data sources + reproducibility

Source	Path
Decision logs	VPS /home/ops/polymarket-5m-bot/state/decisions/2026-05-2{1,2,3}.jsonl (660K+748K+339K bytes)
Bot config (env)	VPS /home/ops/polymarket-5m-bot/.envrc
Strategy code	src/polymarket_bot/strategies/spot_aware_v2_1.py @ 4144302
Branch + HEAD	origin/claude/phase-T-directional-revival @ 4144302
Analysis script	/tmp/soak-audit/analyze.py
Chart generator	scripts/research/phase_T_live_soak_charts.py
PDF builder	pdfs/build_phase_T_live_soak_audit_pdf.py
Phase N forensic (parallel)	pdfs/PHASE_N_FORENSIC_AUDIT.pdf — same pattern at smaller scale
Phase P MMD design	design/P.1-market-mode-detector-integration.md — shadow→gate criteria
Paper-vs-live delta	runbooks/paper-vs-live-delta.md — directional has no live path
Phase T revival prompt	phase-prompts/phase-T-directional-revival-2026-05-20.md — original 6 improvements
MMB parked branch	origin/claude/phase-S-market-maker @ 8fe6007 — MMB code + overview PDF

Reproduce this audit:

1. `scp ops@vps:state/decisions/2026-05-2*.jsonl /tmp/soak-audit/`
2. `python3 /tmp/soak-audit/analyze.py`
3. `uv run --with matplotlib python -u scripts/research/phase_T_live_soak_charts.py`
4. `uv run --with reportlab python -u pdfs/build_phase_T_live_soak_audit_pdf.py`

Authored: 2026-05-23 by Claude Opus 4.7 (1M context). 9 charts, 12 sections, ~20 pages. All numbers cited from the raw decision-log JSONLs as of 11:50 UTC.